



SUPPLEMENTARY INJECTION TECHNIQUES

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3BDS Lecture

By the end of this lecture you should be able to:

- Define the role of accessory local anesthetic techniques in pain control.
- Explain the suitability of supplemental local anesthetic technique in different situations.

Intraosseous Anesthesia

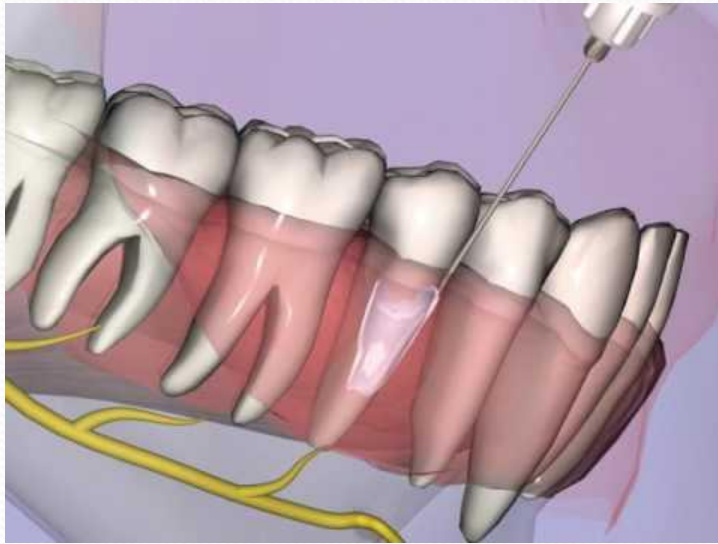
❖ Deposition of anesthetic solution into cancellous bone

1) periodontal ligament Injection
(PDL injection)

2) Intraseptal Injection

3) Intraosseous Injection

Periodontal Ligament Injection



The periodontal ligament
(PDL) injection: an alternative to
inferior alveolar nerve block.

Primary indications for the PDL injection

- (1) the need for anesthesia of but **one or two mandibular teeth** in a quadrant,
- (2) treatment of isolated teeth in both mandibular quadrants (to **avoid bilateral inferior alveolar nerve block**),
- (3) **treatment of children** (because residual soft tissue anesthesia increases the risk of self-inflicted soft tissue injury),
- (4) treatment in which **nerve block anesthesia is contraindicated** (e.g., in hemophiliacs)
- (5) its use as a possible aid in the diagnosis (e.g., localization) of mandibular pain

Hemophilia



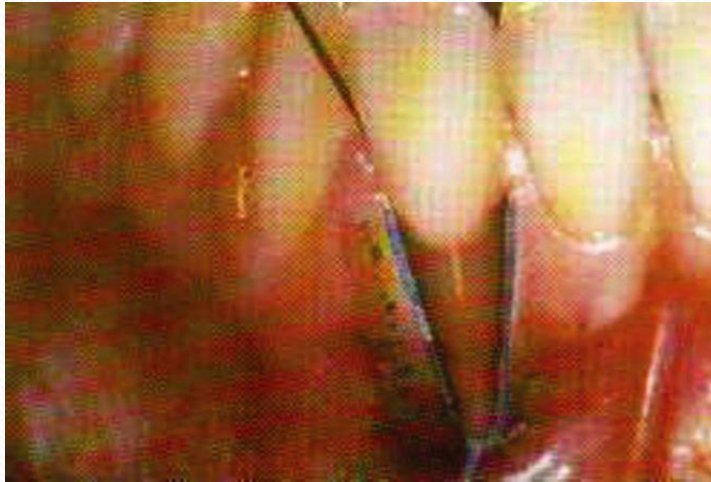
Self inflicted lip injury
after inferior alveolar
nerve block



Contraindications

- 1 Infection or inflammation at the site of injection
- 2 Primary teeth when the permanent tooth bud is present
 - a. Enamel hypoplasia has been reported to occur in a developing permanent tooth when a PDL injection was administered to the primary tooth above it.
 - b. There appears to be little reason for use of the PDL technique in primary teeth because infiltration anesthesia is effective in primary dentition.
- 3 Patient who requires a “numb” sensation for psychological comfort

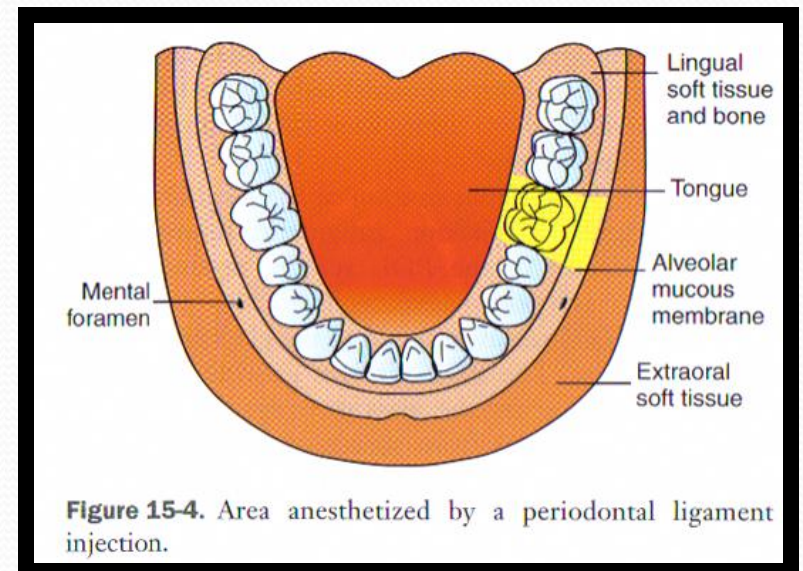
PDL Injection-also known as intraligamentary injection



- ✓ Anesthetic solution reaches the periapical tissues by diffusing apically into the marrow spaces surrounding the teeth into the interseptal bone
- ✓ Solution does not course through the actual PDL to the apex

Periodontal ligament injection

- **Other Common Names**
- Peridental (original name) injection, intraligamentary injection (ILI).
- **Nerves Anesthetized**
- Terminal nerve endings at the site of injection and at the apex of the tooth.
- **Areas Anesthetized**
- Bone, soft tissue, and apical and pulpal tissues in the area of injection



Advantages

1. There is **no anesthesia of the lip, tongue**, and other soft tissues, thus facilitating treatment in multiple quadrants during a single appointment.
2. **Minimum dose of local anesthetic** necessary to achieve anesthesia (0.2 mL per root)
- 3. An **alternative to partially successful** regional nerve block anesthesia
4. **Rapid onset of profound pulpal and soft tissue anesthesia (30 seconds)**
5. Less traumatic than conventional block injections
6. **Well suited for procedures in children**, extractions, and periodontal and endodontic single-tooth and multiple-quadrant procedures

PDL Injection Technique

- 1) 27 gauge short needle is recommended
- 2) Inject on the mesial or distal of the root
- 3) Target area: **depth of the gingival sulcus**
- 4) Orient the bevel toward the root although not important
- 5) May be necessary to bend the needle in order to reach some areas
- 6) Inject down the long axis of the tooth
- 7) Resistance to injection of the solution is a good sign of success
- 8) **Use 0.2 ml** of solution over 20 seconds), Provide the injection for each root on multi-rooted teeth (0.2 ml each)

Area of insertion for a periodontal ligament injection. A, Buccal. B, Lingual.

- Administration of lidocaine with 1 : 100,000 epinephrine, for example, provides pulpal anesthesia ranging in duration from 5 to 55 minutes.



PDL Injection Disadvantages

- Anesthetic solution **leaks to the mouth tastes nasty**
- Needle placement on the **distal of some mandibular molars is difficult**
- Special syringe may be necessary; strength required
- Excessive pressure produces focal tissue damage
- Post-injection discomfort for 2-3 days

Intraseptal Injection

Primary Use - soft tissue and bone anesthesia for periodontal curettage and surgical flap procedures (periodontal surgery)

- Very poor/brief pulpal anesthesia; is not recommended for extractions or for restorative work
- Zero chance of positive aspiration

Intraseptal Injection Technique

- 1) 27 gauge short needle recommended
- 2) Area of insertion: **center of the interdental papilla** adjacent to the tooth being treated
- 3) Inject at the papillary triangle; about 2 mm below the tip; equidistant from adjacent teeth

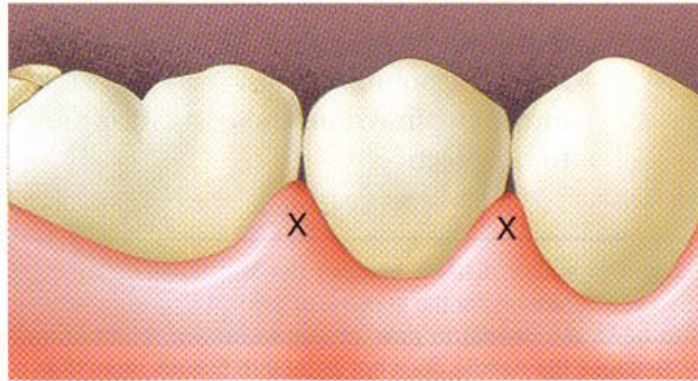


Figure 15-7. Area of insertion for an intraseptal injection.

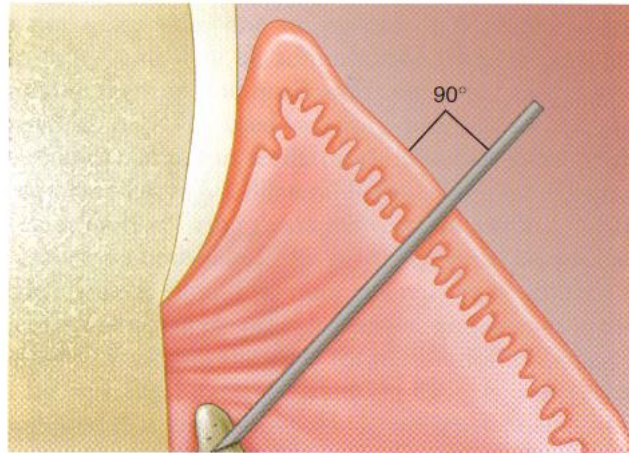
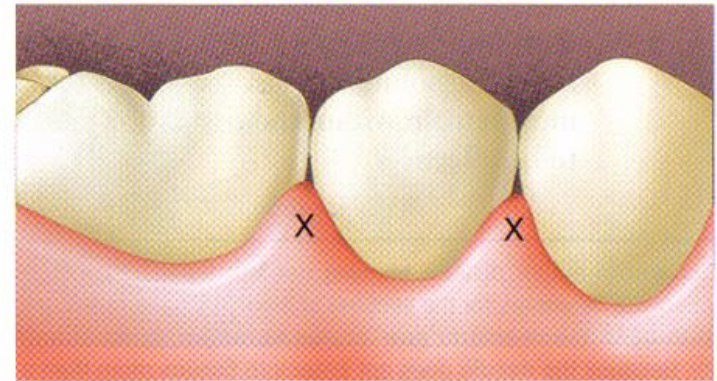
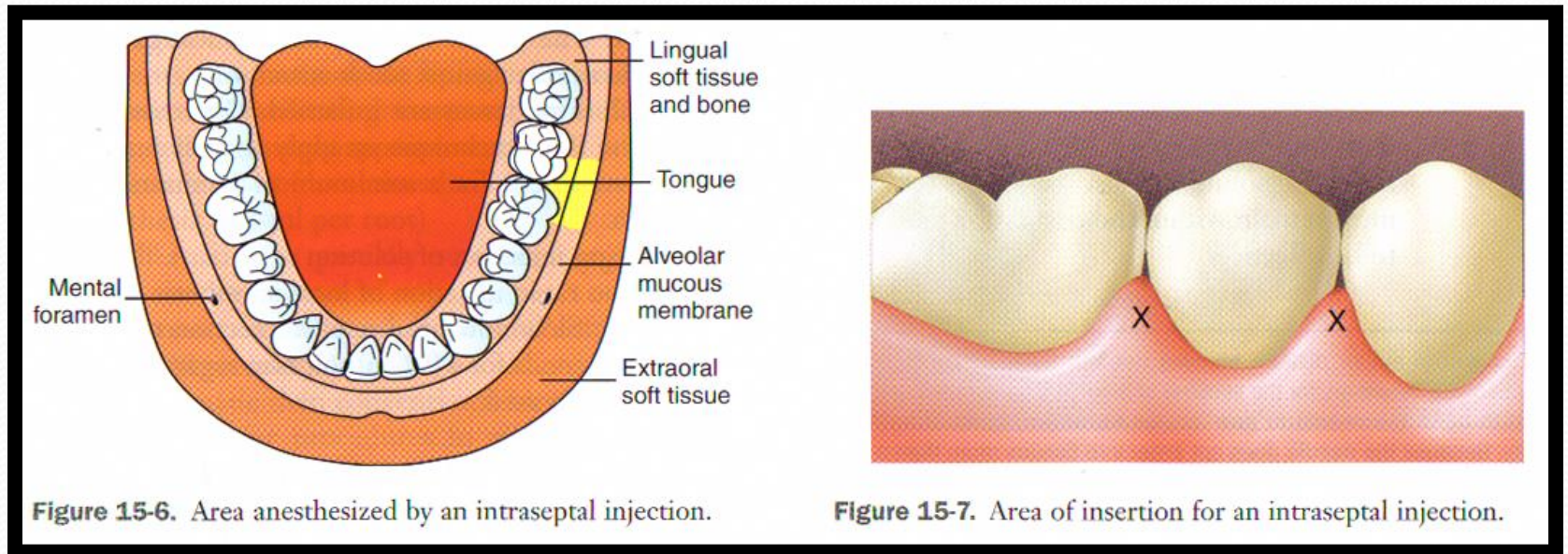


Figure 15-8. Orientation of the needle for an intraseptal injection.

- 4) Enter the gingiva at a 90 degree angle
- 5) Assure needle contacts the bone
- 6) After bone is contacted, apply pressure to the needle to advance it 1-2 mm into the interdental septum (needle tip actually penetrates the bone)
- 7) Deposit 0.2 – 0.4 ml of solution

Areas Anesthetized

Bone, soft tissue, root structure in the area of injection



Intraosseous Injection



The intraosseous injection allows placement of a local anesthetic solution directly into the cancellous bone adjacent to the tooth to be anesthetized

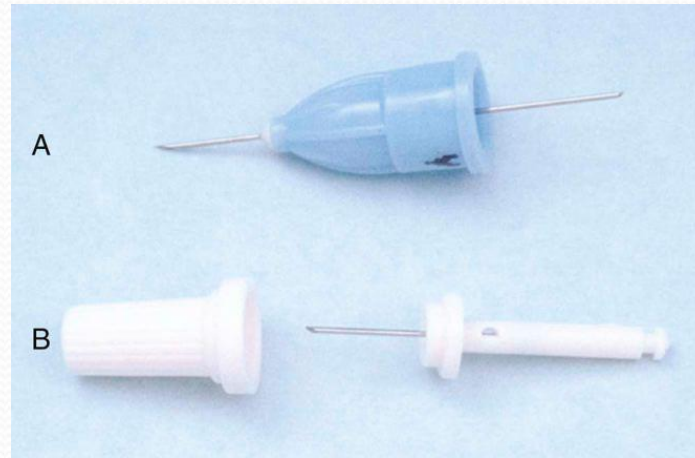
Intraosseous Injection

3 Main Systems:

- 1.) Stabident
 - 2.) X-Tip
 - 3.) Intraflow
- o Generally, there is a perforator (solid needle), 27 gauge needle to deliver the anesthetic and guide sleeve to guide the needle to the pre-drilled hole; perforator is used on a slow speed latch grip handpiece
 - o Inject 0.4 – 0.65 ml into the bone when treating one/two teeth
 - o Inject 1.8 ml when multiple teeth in a quadrant are being treated
 - o Zero chance of aspiration; avoids lip/tongue anesthesia

Intraosseous anesthesia—Stabident.

Components: A, Needle B, Perforator



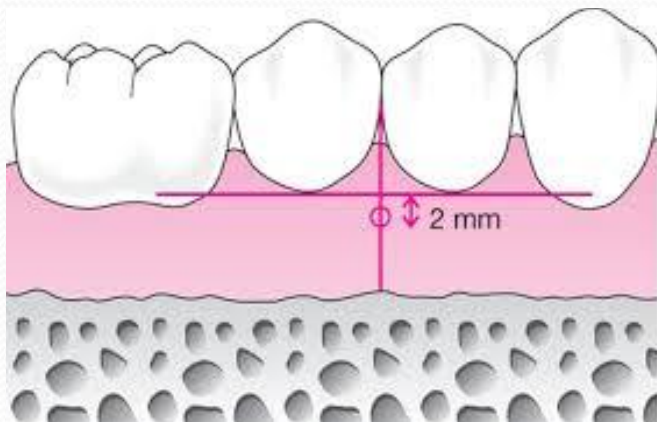
The Stabident® system is comprised of a slow-speed handpiece driven perforator, a solid 27- gauge wire with a beveled end that when activated, drills a small hole through the cortical plate.

The anesthetic solution is delivered to cancellous bone through the 27-gauge ultra-short injector needle placed into the hole made by the perforator.



Intraosseous Injection Technique

- 1) Perforate 2 mm apical to the intersection of lines drawn horizontally along the gingival margins of the teeth and a vertical line through the interdental papilla
- 2) Site should be located distal to the tooth to be treated





1) Hold the perforator perpendicular to the cortical plate of bone

Push the perforator through the soft tissues until bone is contacted

2) Activate the slow speed handpiece penetrating the bone with a pecking motion until resistance is lost

3) Hold the guide sleeve in place until the drill is withdrawn

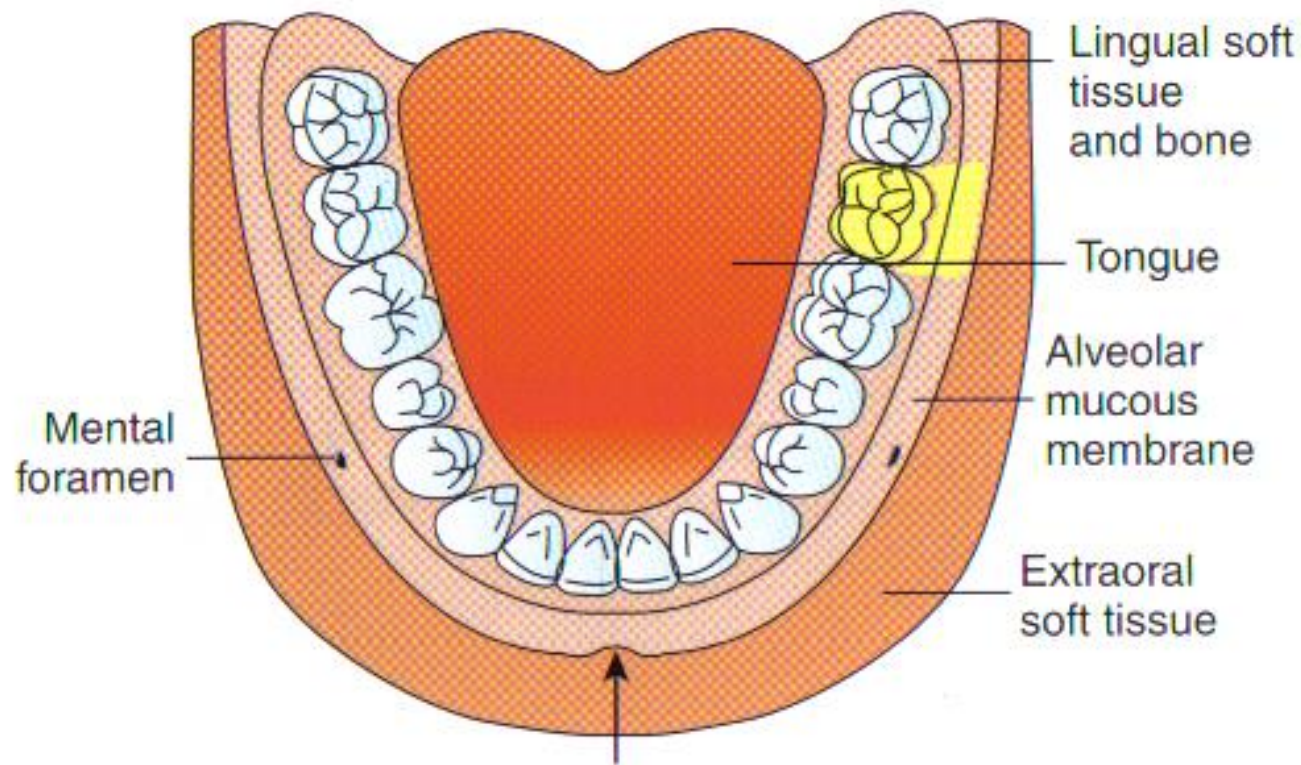
4) The guide sleeve remains in place until you are sure you have adequate anesthesia (if you need to reinject immediately)

5) Insert the needle into hole using a 27 gauge short needle and give anesthesia

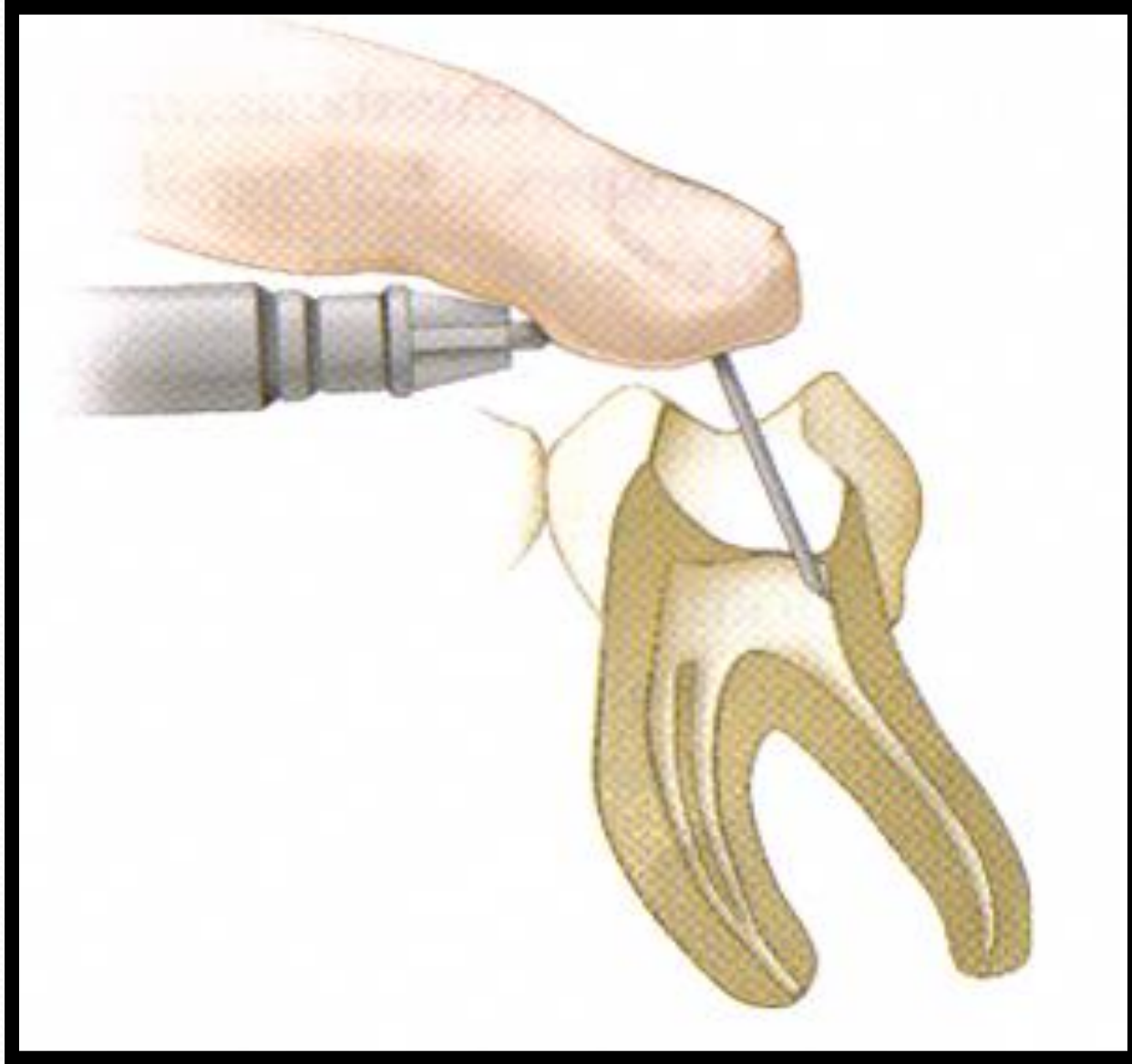
Insert needle into guide sleeve and inject
local anesthetic solution



Areas anesthetized by intraosseous



Intrapulpal Injection



- Used when all other techniques have failed or during endodontic therapy as an adjunct
- Injection is associated with brief pain when injected
- Zero aspiration risk
- Used most commonly on mandibular molars
- Intense, instantaneous pain is usually felt by the patient



Intrapulpal Injection Technique

- 1) 25/27 short or long needle recommended
- 2) May be necessary to bend the needle to gain appropriate access
- 3) Place the needle firmly into the pulp chamber
- 4) Under mild/moderate pressure, deliver the anesthetic solution
- 5) Deposit about 0.2 – 0.3 ml of solution
- 6) Needle breakage could occur after bending the needle but retrieval is rather straightforward
- 7) Begin treatment in 30 seconds; profound, immediate anesthesia
- 8) Slight to severe pain is encountered at times but disappears quickly

Bending The Needle

There are only two injections where bending the needle is acceptable because the needle can be easily retrieved if separated:

- 1) PDL Injection
- 2) Intra pulpal Injection

All other injections can be safely administered without bending the needle whatsoever



The end